

Brazilian E-Commerce by Olist

Final Report

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Introduction

In today's rapidly evolving e-commerce landscape, understanding market trends is crucial for businesses striving to stay competitive. Online marketplaces serve as a bridge between sellers and customers, providing growth opportunities while also facing challenges related to competition, customer satisfaction, and delivery performance.

Olist is one such platform, enabling small and medium-sized businesses to showcase their products and reach a broader customer base. However, the success of sellers depends on several factors, including the type of products they offer, their overall performance in the marketplace, and customer satisfaction regarding delivery experiences. Analyzing these elements is essential for identifying areas of improvement and enhancing both sales and user experience.

This analysis aims to explore the most in-demand products, identify top-performing sellers, and examine the relationship between delivery performance and negative ratings. By understanding these factors, we can develop strategic insights to optimize operations and drive sustainable growth for the platform.

Business Impact

This analysis provides actionable insights to enhance sales and customer experience:

- **Optimizing inventory and marketing** by identifying high-demand products. For example, this helps ensure that top-selling items are always in stock and allows for more effective, targeted marketing campaigns.

- **Strengthening relationships with top sellers** by recognizing their contributions to the platform and offering incentives to encourage continued high performance and reliability.
- **Improving delivery performance** by identifying and addressing delays that negatively impact customer reviews. Timely deliveries are key to ensuring a positive customer experience and encouraging repeat purchases..
- **Enhancing pricing and promotion strategies** by leveraging insights on customer behavior and sales trends.

By implementing these findings, Olist can drive higher sales, improve decision-making, and enhance overall platform efficiency.

Data

File Name:	archive
Description:	Data on orders, products, Sellers, and customers at Olist Store from 2016 to 2018
Dataset Details:	99,441 Order and (9 CSV files with Different lengths)
Size:	42.6 MB (44,717,580 bytes)
Source:	Kaggle - Dataset Link

Data Analysis & Computation

Data Profiling:

- Imported the dataset into an **DuckDb** database using **DBeaver**.
- Examine dataset through **SQL** queries to understand its structure.
- Query essential data relevant to the problem statements.
- Exported the dataset to Excel for further analysis.

- Applied **Excel filters** to identify patterns and inconsistencies.
- Analyzed the unique values of each attribute for irregularities.

Notable Features (Prior to Export)

- **99,441** Unique Order, the order may have more than one seller or product.
- **112,650** Unique items sold.
- **3,095** Unique seller, Large orders for each one.
- **72** Unique product categories, Large orders for each category .
- **27** Unique states and **4,119** Unique cities, with large order per region.
- **There are 8 order statuses:**
 - approved, Canceled, created, delivered, invoiced, processing, shipped, unavailable
- **5 payment types:**
 - boleto, credit_card, debit_card, voucher, not_defined
- compare **review score** with The difference in days from (**order_purchase_timestamp and order_delivered_customer_date**) .
- **Price:** A possible gauge for potential revenues.

Recorded Inconsistency:

- many orders contain **unknown product and seller name and price** (null value).
- Several orders with **no reviews score**.
- many orders contain **unknown product category name** .
- Some values are written in **scientific notation** .
- Some orders have an **unknown order delivery date**.
- One order contains an **unknown payment type (Null value)**.
- One seller has **numerical fields within text strings (seller state)**.
- Exaggerated Customer wait delivery time are suspected outliers and may be test listings.

Data Wrangling:

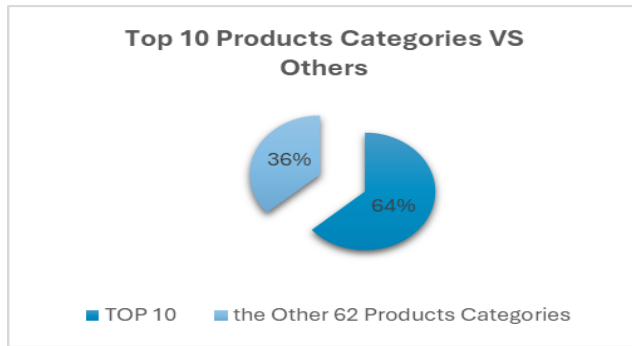
- Using **SQL**, extracted **21 columns** from **9 tables** by applying different JOIN types to merge them into 5 main tables: **Orders, Order Items, Customers, Products, Sellers**.
- The payment_type table was excluded as it is **not relevant** to the analysis.

Column Name	Number of Null Values	Action Made	Reason
product_name	595	Fill it with "unnamed_product"	Maintain data consistency
review_score	646	Fill it with the Mean	Preserve data integrity for analysis
order_delivered_customer_date , price	2,963 , 775	Filtered order_status column to keep only "delivered" (Total removed: 2,963 rows)	Missing delivery dates make it impossible to track orders, and missing prices affect revenue analysis.
order_delivered_customer_date	8	Estimated using average delivery time	Ensure delivery time calculations are complete

- **Number of Unique Columns:** 16, at 5 main tables
- **Number of unique Orders:** 96,478

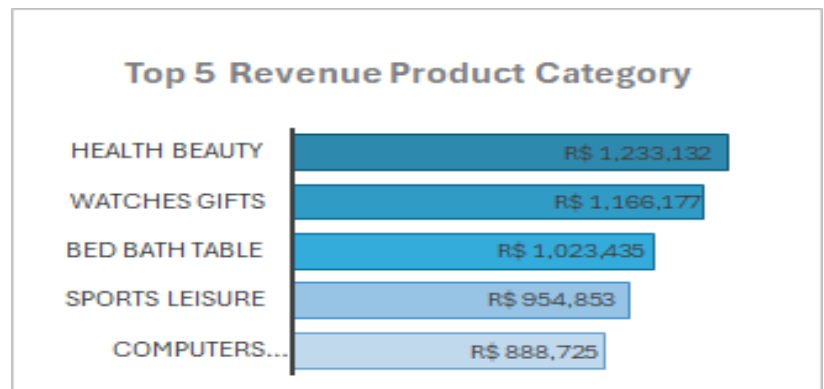
Data Analysis :

- 1- the most requested products



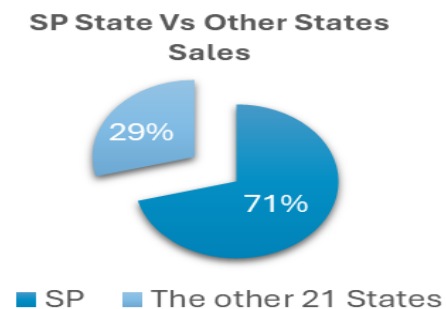
64% of Orders are concentrated only in **14%** (**top 10**) of products Categories, which indicates the concentration of sales on a few popular products. This can be easily visualized using **Pivot Table** and **Pie Chart** through Excel

- Top 5 Products Categories Revenue (**Bar Chart**): “Health Beauty” is the top-performing product category, suggesting a strong customer preference in this sector.

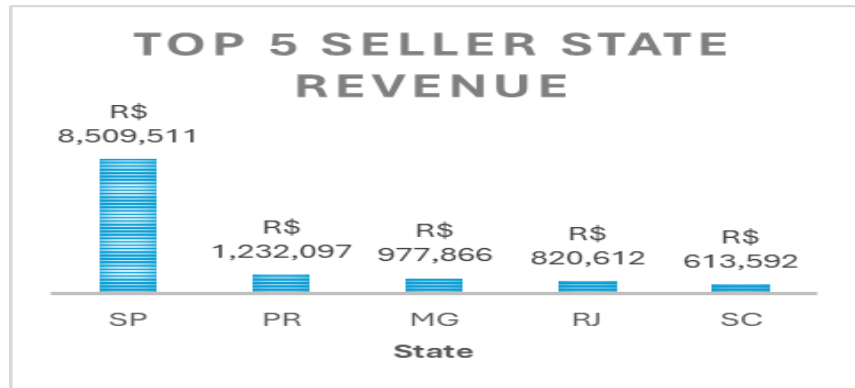


2- States of the most active sellers and their impact on Sales

- 71%** of Orders are concentrated only in SP State, which indicates weak sales in other states. This can be easily visualized using **Pivot Table** and **Pie Chart** through Excel.

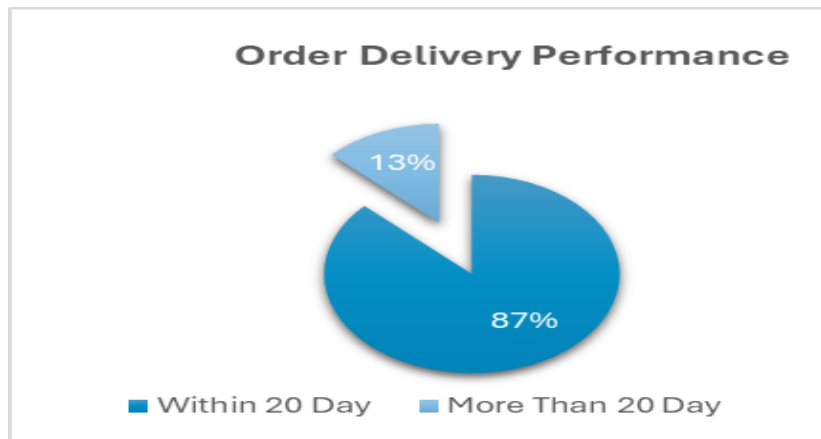


- Top 5 Seller State Revenue (**Bar Chart**): There's a massive gap in revenue between states, with São Paulo (SP) generating 64% of the revenue



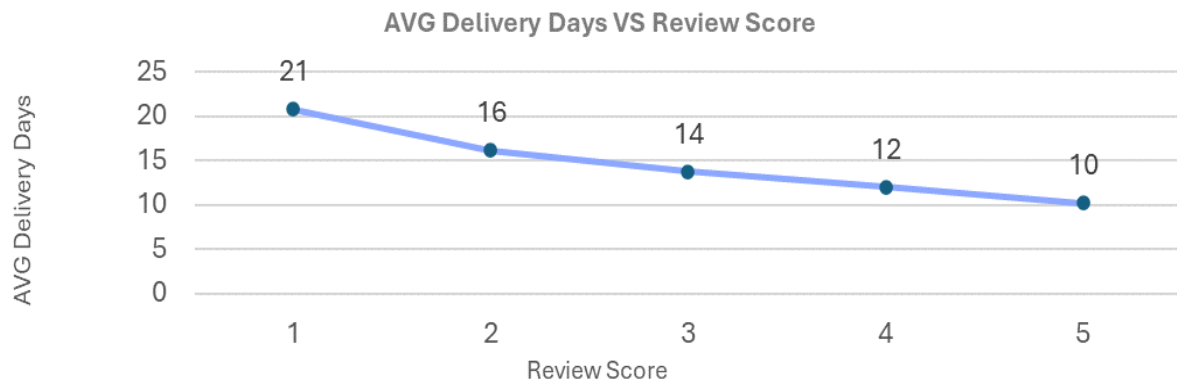
3- delivery performance and its relationship to Review Score

- **87%** of orders were delivered Within 20 days, and **13%** exceeded this period, which indicates that delivery performance is good. This can be easily visualized using **Pivot Table** and **Pie Chart** through Excel.



- The relationship between Delivery Days and Review Score

- There is a negative correlation between the number of Delivery Days and Review Score, meaning that faster deliveries tend to receive higher review scores the correlation coefficient (CORREL () = - 0.3)
- The Line Chart shows the relationship between the Average of Delivery Days and Review Score, lower delivery days leads to an increase in the review score



4- Basic Statistics

- Descriptive statistics were used to understand the distribution of prices, ratings, and number of orders.

measure	price	review score	# of orders by state
mean	124	4	39
median	79	4	8
St. Div	188	1	126
upper bound	286.75	5	63
lower bound	-107.25	1	-33
skewness	7.8990	-1.4580	9.6955

Conclusion & Future Work

Conclusion:

The analysis conducted has provided valuable insights into product demand, sales distribution, and delivery performance. The results showed that 64% of orders are concentrated in only 14% of product categories, highlighting high demand for certain products. Additionally, it was found that SP State contributes 71% of the total number of orders, indicating the significant impact of geographic location on order volume. Faster deliveries lead to higher customer satisfaction,

while delays negatively affect review scores. There is also a significant opportunity to improve marketing and pricing strategies to increase revenue.

Future Work:

While the analysis has provided valuable insights into market performance, there are areas for improvement, including:

- **Geographical Market Expansion:** Collecting more granular geographic data to understand how location impacts orders, particularly in underperforming regions.
- **Payment Method Analysis:** Studying the impact of different payment methods on order delivery and transaction success.
- **Analysis of Unfulfilled Orders:** Investigating the reasons behind delayed or unfulfilled orders and working to improve these processes.

Further research in these areas will help refine current strategies and fully leverage the market's potential.